

# Estimation of gas puff Z-pinch parameters by using snowplow model simulation and the $\beta$ factor variation

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**Abstract:** The gas puff z-pinch device is used to investigate the dynamics of injected plasma by neutral particles; during the injection of the plasma by the neutral particles, the temperature is lowered and the plasma density is increased. The snowplow model is used to simulate the gas puff z-pinch device and estimate the gas puff pinch parameters. The model takes into account the radial compression, the reflected shock wave, and the expansion phases. The electric circuit and momentum equations are used and solved numerically by the model. Furthermore, the relationship between the kinetic pressure and the  $\beta$  factor, which ranges from 0 to 0.9, is studied in the model. The model demonstrates how increasing the  $\beta$  factor and injected neutral gas reduces temperature and velocity. The pinching time is increased with increasing the  $\beta$  factor and reduced with increasing the injection time.

**Keywords:** Gas puff pinch;  $\beta$  factor; Snowplow model; Plasma radius; Pinching time; Injection time

## Introduction

The concept of gas puff Z-pinch is widely used; a hollow gas puff Z-pinch can be produced by injecting an annular gas jet, usually from the cathode to the anode. The plasma dynamics obtained from Schlieren images are compared to the predictions of a 2-D MHD Lagrangian code. Numerical calculations illustrate that the thickness variations along the axis and axial distribution of the linear density are controlled in the behavior of the gas puff z-pinch [1]. The goal of the gas-puff Z-pinch experiment is to study the dynamic behavior of the implosion for two gas-puff Z- $\theta$  pinch compressions [2] and to gain an understanding of the dynamics and instability development in the imploding and stagnating plasma, where the experiment is operated on the 1 MA, 200 ns pulse duration [3]. The plasma dynamics are described experimentally by the coaxial plasma device with gas puff in a low-pressure environment and theoretically by the snowplow model, where the ionization of an axially propagating plasma is described by high-speed camera images [4].

The temperature and density of the fast z-pinch system are 50 eV and  $1.4 \times 10^{26} \text{ m}^{-3}$ , respectively, and it is

experimentally investigated using injected neon and argon gases to generate ultra-soft X-rays using a high current pulse generator of 10  $\mu\text{F}$ , 10 kJ [5]. However, the hot spots at the pinch point, where the temperature and density are approximately 1 keV and  $10^{21} \text{ cm}^{-3}$ , respectively, are investigated using a crystal spectrometer, which can be a good source for spectroscopy and X-rays [6]. The hot spot distribution and x-ray radiation are investigated by changing the gas nozzle shape, where the temperature and intense x-ray emission are greater, reducing the flux of the gas [7].

The large-diameter gas-puff Z pinch is designed to generate high-yield K-shell radiation, where the Ar K-shell yield at 3.46 MA is increased in 200 ns implosions from 12 cm initial diameter by a factor of 2 to 21 kJ [8]. The relation between the radial pinching ratio and the current flowing magnitude at the plasma is studied, where the plasma radius in the K-shell lines of neon gas is about 3–4 mm; therefore, the radiation yield of K-shell lines is about 5–11 kJ/cm, and then the major part of the current flows through the plasma column (skin effect), while the minor part of the current flows through the central plasma region [9]. A triple-nozzle gas-puff z-pinch with 1 MA and 220 ns are used to study the implosion dynamics of plasma, where the stability and dynamics of plasma implosion are compared with analytical models and experimental data. The snowplow models and Classical Potter are given an

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accurate description of radial implosion trajectories. Different gas species and axial magnetic field strengths are used to observe a difference in the structure of the accelerating plasma sheath and evolution of the MRT instability [10].

The magnetic compression of cylindrical liners filled with DT gas is used to achieve nuclear fusion by using a pulsed system to avoid rapid cooling of the fuel by transfer of heat to the liner [11] and to insulate the hot compressed plasma from the cold liner [12]. Using an external axial magnetic field and XUV in a gas puff z-pinch device with a Kr liner and a D target, the stability of the outer plasma is determined using 1 MA, 100 ns, where the external axial magnetic field reduces the total instability amplitude. [13]. The gas flow doesn't distribute symmetrically about  $r_0$  in the supersonic nozzle gas puff device, where it is found that, at a gas pressure of 5 atm., the measured line mass density is 43 g/cm, and the Mach number is deduced from experiments of 4.3 and a line mass density of about 50 g/cm [14].

The snowplow model was used to simulate both theta pinch [15] and hollow cathode [16] devices. The variation impacts of the  $\beta$  factor, which equals the kinetic pressure to the magnetic pressure ratio and applied current in the theta pinch device on the normalized radius of the CS, were studied. The simulated results showed that with decreasing  $\beta$  value and increasing applied current, the theta pinching effect can be improved [15]. The impact of the magnetic pressure, shock wave pressure, and kinetic pressure as a function of the  $\beta$  factor was studied in the hollow cathode. By decreasing the  $\beta$  value, the propelled plasma velocity increases [16]. The Magnetic Pinch Lab at the Plasma and Nuclear Fusion Dep.-Egyptian Atomic Energy Authority aims to simulate a gas puff Z-pinch system to generate the X-ray and to inject a test gas (only second gas) inside the core of the fusion reactor to cool it. Our current studies use the snowplow model, which has to be updated with new terms in order to keep the plasma radius from going to zero. According to this mathematical model, kinetic pressure can have a significant impact; therefore, the momentum equation of the model is modified by including the shock wave and gas kinetic pressure as a function of the  $\beta$  factor (the  $\beta$  factor is the ratio of kinetic pressure to magnetic pressure). The  $\beta$  factor, or kinetic pressure, is studied by the model to prevent the plasma radius from reaching zero. The comparisons and projections with and without gas puff pinch are made in the next sections.

## Equations of the model

### Circuit equation

The capacitor bank discharges between the two electrodes to generate a current " $I_z$ " in the z-direction, which generates an induced magnetic field " $B_\theta$ " in the  $\theta$ -direction. Based on Fig. 1, the circuit equation can be derived using Kirchhoff's Law [15–18] as:

$$V_0 - \int \frac{Idt}{C_0} = \frac{dI}{dt} (L_0 + L_p f_c) + I f_c \frac{dL_p}{dt} + r_c I \quad (1)$$

The plasma inductance  $L_{p1}$  before the test gas is injected could be written as:

$$L_{p1} = \frac{\mu z_0}{2\pi} \ln\left(\frac{r_0}{r_1}\right)$$

But the plasma inductance  $L_{p2}$  after the test gas is injected could be written as:

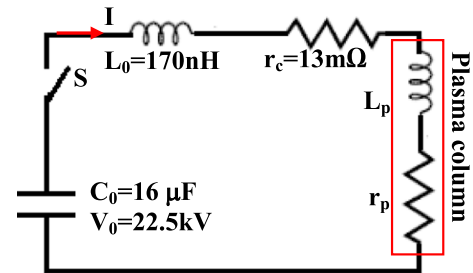
$$L_{p2} = L_{p1} + \frac{\mu z_0}{2\pi} \ln\left(\frac{r_1}{r_2}\right)$$

where  $C_0$ ,  $L_0$ ,  $I$ ,  $r_c$ ,  $V_0$ ,  $r_0$ ,  $r_1$ ,  $r_2$ ,  $z_0$ ,  $\mu$  and  $f_c$  are the capacitance of the capacitor, the circuit inductance, discharge current, the circuit resistance, the charging voltage, the electrode radius, the plasma radius before the test gas is injected, the plasma radius after the test gas is injected, the distance between the two electrodes, the permeability of the medium, and the current factor, which is used to obtain the best fitting of the simulated and experimental current signals. The current equation can be given by:

$$\frac{dI}{dt} = \left( V_0 - \int \frac{Idt}{C_0} - I r_c - I f_c \frac{dL_{p2}}{dt} \right) / (L_0 + L_{p2} f_c) \quad (2)$$

### Equation of motion

The plasma gas puff pinch device is simulated using the snowplow model. The induced magnetic field " $B_\theta$ " in the  $\theta$ -direction is generated when a capacitor discharges



**Fig. 1** Schematic diagram of the equivalent electric circuit of gas puff pinch device that includes the condenser bank, power supply and switch

between the two electric electrodes, generating a discharge current “ $I_z$ ” in the z-direction. So, the major magnetic force “ $F_{Br}$ ” is generated in the r-direction when the  $I_z$  and  $B_\theta$  are perpendicular; therefore, this force compresses the current sheath (CS) inwardly towards the axis as shown in the left side of Fig. 2. Two gases are used inside the discharge chamber of the gas puff pinch mechanism. The first and second gas densities are assumed by the model to be  $\rho_1$  and  $\rho_2$ , respectively. The first one is flowed through the tube to create and generate the CS with radius  $r_1$ , while the second one is injected by high velocity through the plasma center with radius  $r_2$ . According to the model, the electrode radius is indicated by “ $r_0$ ,” and as shown in the right side of Fig. 2,  $r_2 < r_1 < r_0$ . The magnetic force  $F_{Br}$ , the shock wave force  $F_{sr}$ , and the kinetic force  $-F_{kr}$  make up the total force  $F_{Fr}$  that the model uses to compress the plasma inwardly towards the axis. Based on the shock wave theory, the shock wave is created when the CS speed exceeds the supersonic speed, so the shock wave advances ahead of the CS to ionize and sweep the neutral particles, and then a slug of gas forms between the CS and SF as illustrated in Fig. 2.

This slug is forced to compress by the shock wave, and the plasma is pushed outward and radially by the kinetic force  $-F_{kr}$ . The motion equation can be given in the snowplow model by [16, 20–23]:

$$\frac{d}{dt} \left( M \frac{dr_p}{dt} \right) = F_{Fr}, \text{ But } F_{Fr} = F_{Br} - F_{kr} - F_{sr} \quad (3)$$

where  $dr_p/dt$  and  $M$  are the radial plasma velocity and the plasma mass. The plasma mass can be calculated by  $M = \rho \cdot V$ , where  $V$  is the total volume, which equals  $V = \pi z_0(r_0^2 - r_1^2)$ , so the total mass of plasma in the gas puff pinch device can be divided into two parts: the initial gas and test gas masses, which can be given by:

$$M = \rho_1 V_1 (V_1 \text{ is the volume of the initial gas}) + \rho_2 V_2 (V_2 \text{ is the volume of the test gas})$$

$$M = \pi z_0 f_m \rho_1 (r_0^2 - r_1^2) \text{ volume of the initial gas} + \pi z_0 f_m \rho_2 (r_1^2 - r_2^2) \text{ volume of the test gas}$$

$$M = \pi z_0 f_m [\rho_1 (r_0^2 - r_1^2) + \rho_2 (r_1^2 - r_2^2)] \quad (4)$$

The momentum equation becomes [24]:

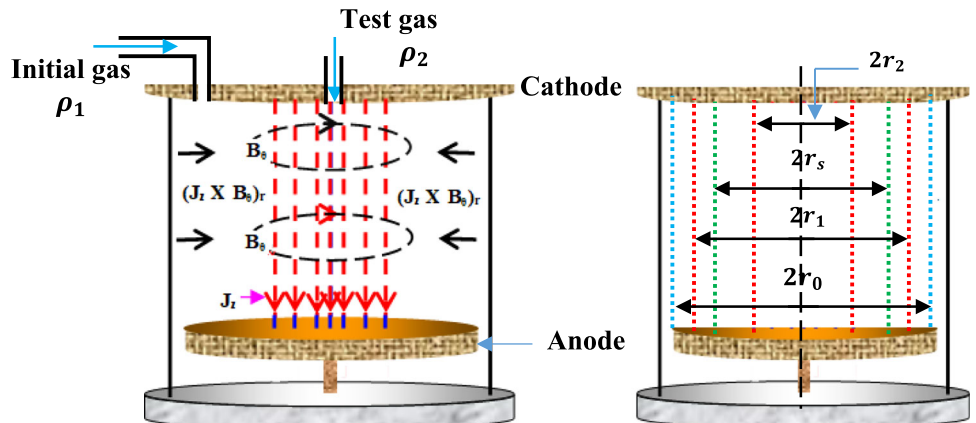
$$F_{Fr} = \frac{d}{dt} \left( M \frac{dr_p}{dt} \right) = \frac{d}{dt} \left( \pi z_0 f_m [\rho_1 (r_0^2 - r_1^2) + \rho_2 (r_1^2 - r_2^2)] \frac{dr_p}{dt} \right) \quad (5)$$

where  $f_m$  is the fraction of mass swept by the current sheath motion. Because the swept mass of the theoretical snowplow model is larger than the actual one, the mass factor  $f_m$  is utilized to lessen the discrepancy between the tiny values of the estimated velocity and the actual measured experimentally [25]. The total force  $F_{Fr}$  is deduced as:

$$F_{Fr} = ((P_{Br} - P_{kr}) - P_{sr}) A_p = \left( P_{Br} \left( 1 - \frac{P_{kr}}{P_{Br}} \right) - P_{sr} \right) A_p \quad (6)$$

where  $P_{Br}$ ,  $P_{kr}$ ,  $P_{sr}$ , and  $A_p$  are the magnetic, kinetic, shock wave pressures and the area of the plasma, which is compressed by the force  $F_{Fr}$ . The  $\beta$  factor (the ratio of the kinetic pressure to the magnetic pressure) can be written as  $\beta = \frac{P_{kr}}{P_{Br}}$  so,  $F_{Fr} = (P_{Br}(1 - \beta) \cdot A_p - P_{sr} \cdot A_p)$ . The model suggests that  $P_{Br,kr} = P_{Br} \cdot (1 - \beta)$ , where  $P_{Br,kr}$  represents the magnetic and kinetic pressures, so the total force  $F_{Fr}$  becomes  $F_{Fr} = (P_{Br,kr} \cdot A_p) - (P_{sr} \cdot A_p)$ . The magnetic pressure can be calculated by  $P_{Br} = \frac{B_\theta^2}{2\mu}$ , where  $B_\theta$  is the azimuthal magnetic field intensity, which is given by  $B_\theta = \frac{\mu I_z f_c}{2\pi r_{1,2}}$ , and then  $P_{Br} = \frac{\mu I_z^2 f_c^2}{8\pi^2 r_{1,2}^2}$ . The final magnetic and kinetic pressures relation can be given as:

**Fig. 2** The CS and SF dynamics inside the gas puff pinch device with injected test gas



$$P_{Br,kr} = P_{Br}(1 - \beta) = \frac{\mu I_z^2 f_c^2 (1 - \beta)}{8\pi^2 r^2}$$

Then, the final magnetic and kinetic forces can be given by:

$$F_{Br,kr} = \int_{r_2}^{r_0} P_{Br,kr} \cdot dA_p = \frac{\mu f_c^2 (1 - \beta) I_z^2}{2\pi} \ln\left(\frac{r_0}{r_2}\right) \quad (7)$$

Additionally, the model makes use of the shock wave, which advances ahead of the current sheath CS until it reaches the axis and reflects back towards it, while the current sheath continues to advance towards the axis to meet it at the pinching point [16, 22–25], so the shock force can be given by:

$$F_{sr} = P_{sr} \cdot A_p = \left[ \frac{4\pi(\rho_1 + \rho_2)f_m}{\gamma + 1} \right] r_1^2 V_s^2 \quad (8)$$

where  $P_{sr} = 2(\rho_1 + \rho_2)V_s^2/(\gamma + 1)$  is the shock wave pressure,  $V_s$  is the shock wave velocity [18] and  $\gamma$  is the specific heat ratio for ideal gas [26], so the final equation of motion can be written as:

$$\frac{d^2 r_p}{dt^2} = \frac{\alpha I_z^2 \ln\left(\frac{r_0}{r_2}\right) - \delta r_1^2 V_s^2 + \sigma(2r_1 V_1^2(\rho_1 - \rho_2) - 2\rho_2 r_2 V_1 V_2)}{\sigma(\rho_1(r_0^2 - r_1^2) + \rho_2(r_1^2 - r_2^2))} \quad (9)$$

$$\alpha = \frac{\mu(1 - \beta)f_c^2}{2\pi}, \delta = \frac{4\pi(\rho_1 + \rho_2)f_m}{\gamma + 1} \text{ and } \sigma = \rho_1 \pi f_m z_0$$

Where  $V_1$  and  $V_2$  are the plasma velocity before and after gas puffing, according to the thermal dynamics, the shock wave velocity that moves ahead of the current sheath to sweep the neutral particles can be estimated by [19, 27, 28]:

$$V_s = \left[ \frac{\gamma + 1}{2} \right] \frac{dr_p}{dt} \quad (10)$$

Plasma temperature

The plasma temperature is calculated by the model, where it is dependent on the shock wave velocity. The plasma temperature equation is provided by [29, 30]:

$$T = \frac{2M_g(\gamma - 1)}{R_0 D(\gamma + 1)^2} V_s^2 \quad (11)$$

where  $M_g$  is the gas molecule weight,  $R_0$  is the universal gas constant, and  $D$  is the departure coefficient;  $D = D_N(1 + Z_{eff})$ ,  $D_N$  is the gas dissociation number,  $Z_{eff}$  is the effective charge state [20, 21].

## Results and discussion

### Plasma properties before and after gas puff

This paper will study the injection of neutral particles by the gas puff device using the snowplow model simulation. The effect of the  $\beta$  factor on plasma properties is also estimated. The model will be operated under the following conditions (Table 1):

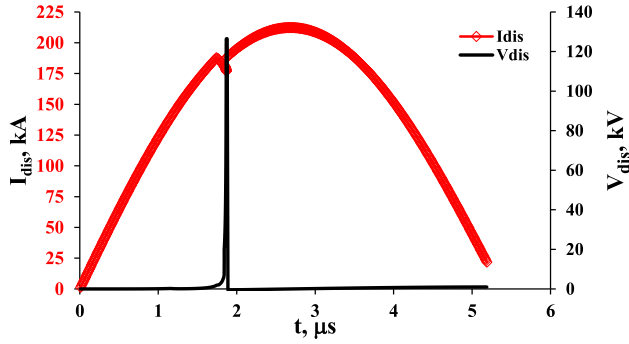
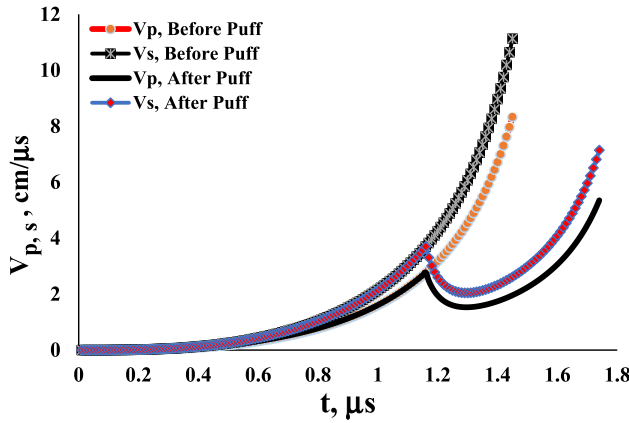
The model is run by using neon gas (test gas) that is injected into the center of the plasma column at different times in order to estimate both the discharge current, discharge voltage, plasma velocity, temperature, and density. The capacitor's capacitance is 16  $\mu$ F, and the charging voltage is 22.5 kV, respectively [31]. Figure 3 shows how the discharge current  $I_{dis}$  rises quickly over a 2.6  $\mu$ s period. At a time of around 1.8  $\mu$ s, a crevice in the discharge current signal and a spike in the discharge voltage  $V_{dis}$  signal appear, indicating that the plasma is compresses and squeezes towards the axis, where the magnetic pressure exceeds the kinetic pressure [32]. The mass factor  $f_m$  and the current factor  $f_c$ , which typically range between 0.1, 0.33 and 0.66, 0.8, respectively, are employed in various devices to ensure that the simulated and experimental data match [19, 33, 34]. The current paper uses different values for  $f_m$  and  $f_c$ , which are 0.012 and 0.3, respectively [35].

The SF ( $V_s$ ) and CS ( $V_p$ ) velocities in the compression phase are depicted in Fig. 4 with and without the gas puff. Because of increasing the discharge current, the  $V_p$  and  $V_s$  increase with discharge time, but according to Eq. 10, the SF velocity exceeds the CS velocity by  $(\gamma + 1)/2$  [36]. As can be seen from Eq. 10, the  $V_s$  arrives at a maximum value of 11.2 cm/ $\mu$ s without gas puff and 7.1 cm/ $\mu$ s with gas puff [16]; it is evident that the  $V_s$  is bigger than the  $V_p$ , where the test gas is injected into the plasma core, so both  $V_s$  and  $V_p$  decrease, as seen by the difference between the two cases. When a neutral gas (test neon gas) is injected into the plasma column, its mass and density increase, causing the plasma velocity (total mass of two gases; refer to Eq. 4) to decrease at the time of 1.16  $\mu$ s. Subsequently, a thermalization process occurs between the ionized particles and the neutral injected particles, so the injected gas undergoes ionization and compression by shock wave and magnetic pressure, leading to an increase in plasma velocity once more.

The shock wave's radial position  $r_s$  and the plasma's  $r_p$  with and without the gas puff pinch are depicted in Fig. 5 through the radial and reflected phases. At the start of the discharge, the shock wave and plasma radii are larger. The magnetic pressure then causes them to decrease radially in the direction of the axis. At 1.45  $\mu$ s, the shock wave reaches the axis to hit it and reflects to meet the CS, which continues to advance toward the axis at the pinching point

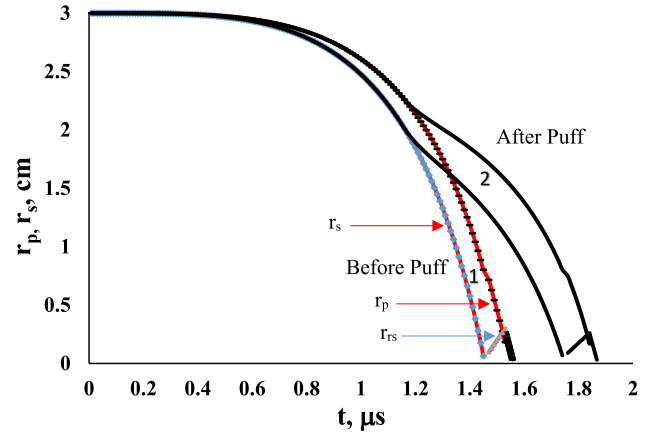
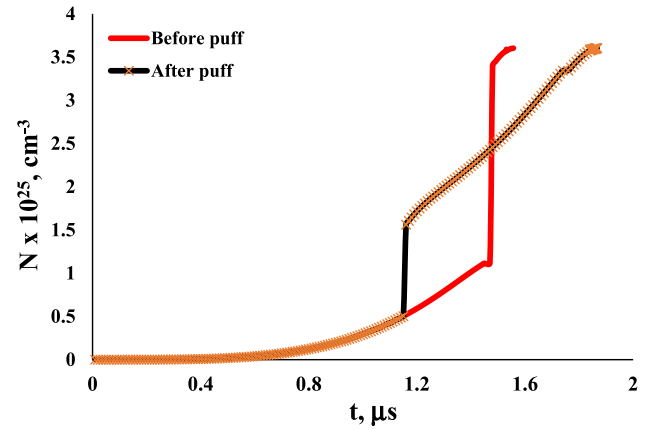
**Table 1** The primary conditions of the model [31]

| Element | $L_0$ | $C_0$   | $V_0$ | $P_1$ | $P_2$ | $r_c$      | $r_0$ | $z$ | $f_m$ | $f_c$ | Gas |
|---------|-------|---------|-------|-------|-------|------------|-------|-----|-------|-------|-----|
| Unit    | nH    | $\mu$ F | kV    | torr  | torr  | m $\Omega$ | cm    | cm  |       |       |     |
| Value   | 170   | 16      | 22.5  | 10    | 3     | 0.013      | 3     | 3   | 0.3   | 0.02  | Ne  |


**Fig. 3** The simulated discharge current  $I_{dis}$  and discharge voltage  $V_{dis}$  as a function of time at 22.5 kV

**Fig. 4** The plasma  $V_p$  and the shock wave  $V_s$  velocities before and after the gas puffing at 22.5 kV

of 1.52  $\mu$ s without a gas puff (Group<sub>1</sub>). At 1.74  $\mu$ s, the shock wave reaches the axis and reflects to meet the CS at 1.83  $\mu$ s with a gas puff (Group<sub>2</sub>). Group<sub>1</sub> and Group<sub>2</sub> in Fig. 5 illustrate the shock wave and plasma radii without gas puff and with gas puff, respectively. As demonstrated by Eq. 4, it is evident that Group<sub>2</sub> requires more time to reach the pinch point, where the kinetic pressure and hence the  $\beta$  factor exceed that of Group<sub>1</sub> due to the larger mass of the CS and its density.

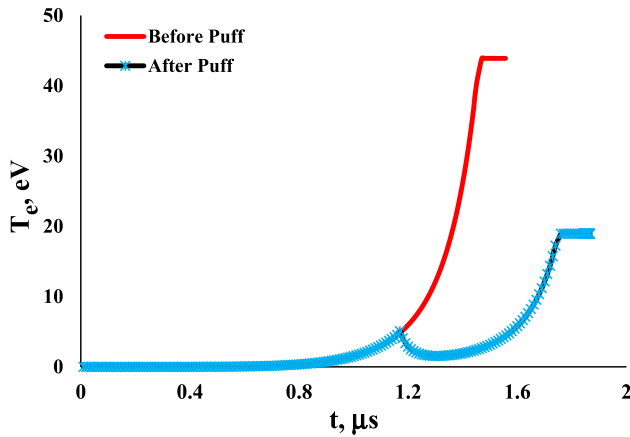
The plasma density  $N$  with and without gas puff pinch is shown in Fig. 6. The plasma density in the absence of gas puff increases gradually with time, peaking at 1.47  $\mu$ s of pinch time [37]. The plasma density increases sharply at 1.15  $\mu$ s and continues to rise when the neutral gas (test gas)


**Fig. 5** The plasma  $r_p$ , the shock wave  $r_s$  and reflected shock wave  $r_{rs}$  radii before and after the gas puffing at 22.5 kV

**Fig. 6** The simulated plasma density ( $N$ ) before and after the gas puffing at 22.5 kV

is injected into the center of the plasma column; as a result, it peaks at 1.84  $\mu$ s of pinch time. Increasing the plasma density occurs by collecting more ions through the sheath motion in the radial direction and also by reaching higher charge states as a result of the higher plasma temperature, as shown in Fig. 6, which causes more ionization processes [36].

Figure 7 shows the plasma temperature  $T$  as a function of time in the radial and reflected phases with and without the gas puff pinch. The temperature of the plasma without the gas puff rises in the direction of the axis, reaching a





**Fig. 7** The simulated plasma temperature  $T$  before and after the gas puffing at 22.5 kV

maximum value of 44 eV. As a result of the shock wave ionizing the neutral gas in front of the plasma sheath, the temperature will rise as a result of collisions between the plasma particles and the ionized particles [16]. The temperature is raised until the shock wave hits the axis, at which point the reflected shock phase begins, as the temperature is dependent on the shock wave velocity as stated in Eq. 11. As seen in Fig. 7, the temperature has a constant value during this phase (reflected phase), where the reflected shock wave velocity has a constant value of 0.3 times the SF velocity at the radial phase end. When the neutral gas is injected into the plasma column during a gas puff, the temperature drops sharply as a thermalization process takes place between the neutral particles and the ionized gas, and then the CS temperature tries to subsequently rise again, as seen in Fig. 7.

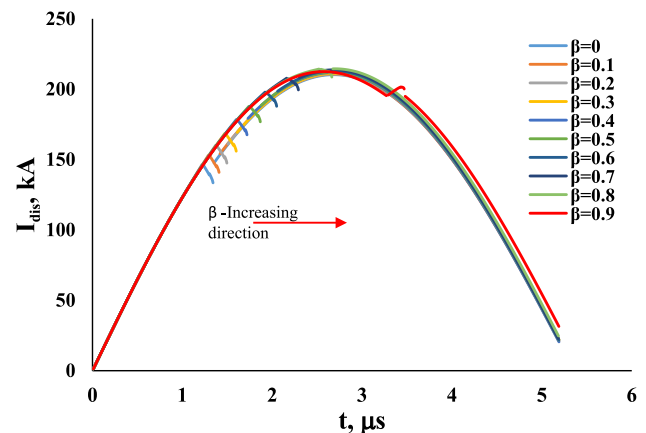
#### Effects of $\beta$ factor variation on the gas puff pinch

The snowplow model is used in the design and operation of the gas puff pinch model in order to simulate the gas puff device. In the normal snowplow model, the kinetic pressure is not taken into account. Instead, the plasma column is compressed and squeezed by the only magnetic pressure in order to reach zero radius. The plasma radius is lowered until it approaches zero since the snowplow model only takes the magnetic pressure into account and ignores the kinetic pressure. However, the kinetic pressure is very effective in gas puff pinch devices, so in the current model, the kinetic pressure is added to the momentum equation as a function of the  $\beta$  factor (the ratio of the kinetic pressure to the magnetic pressure) in order to prevent the plasma radius from reaching zero and study the effects of the kinetic pressure on the plasma gas puff [33]. By using Eq. 9, the model utilizes the  $\beta$  values, which range from 0 to 0.9. The simulated discharge current with the  $\beta$  factor

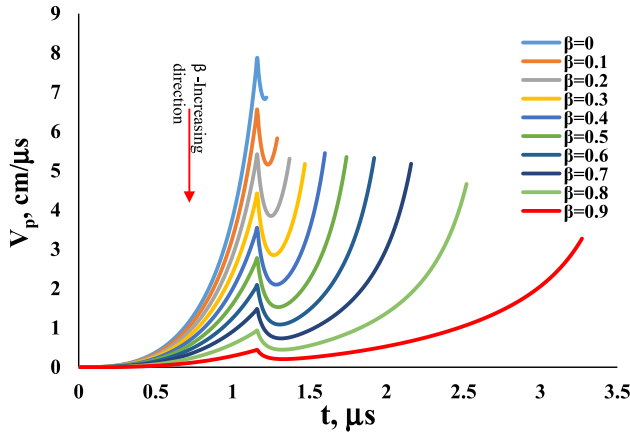
variation as a function of time is displayed in Fig. 8. At various  $\beta$  values, the discharge current grows from zero to a maximum value. Because of the reduction and increase in kinetic pressure as a function of the  $\beta$  factor, there is a clear crevice in the discharge current signal that shows that the plasma column compresses in the radial direction at various times. The figure indicates that, while the kinetic pressure rises with gas puff, the pinching time in the discharge current signal increases with the  $\beta$  factor.

The plasma velocity with the variation of the  $\beta$  factor as a function of time is shown in Fig. 9. Both the  $\beta$  factor and gas puff pinch cause the velocity to decrease as the kinetic pressure rises. The CS and shock wave sweep and ionize the neutral gas particles injected inside the plasma column, affecting the velocity due to a reversal kinetic pressure at the plasma's core. According to the snowplow model, the plasma column suffers a heavy mass, and as a result, Eq. 9 shows that the acceleration is subsequently decreased by an increase in the  $\beta$  factor [35]. Also, it is clear that the CS velocity at  $\beta = 0$  with the gas puff reaches a maximum value in a short time, while at  $\beta = 0.9$  with the gas puff, the velocity needs a longer time of about 3.3  $\mu$ s to reach a maximum value.

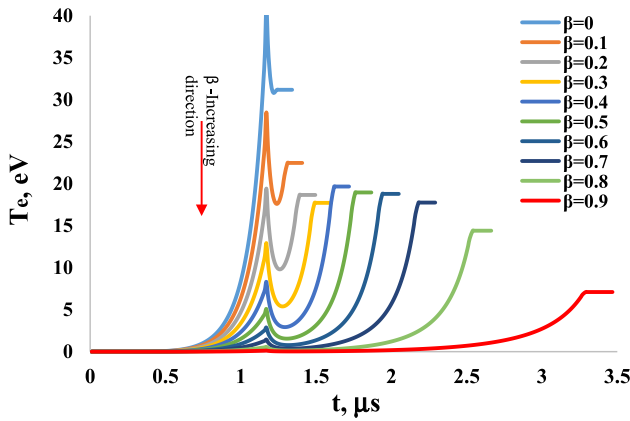
The plasma temperature with the variation of the  $\beta$  factor as a function of the time with gas puff is shown in Fig. 10. When the plasma column is compressed and pinched in the radial direction, the temperature rises with time. At  $\beta = 0$ , the temperature is higher, while at  $\beta = 0.9$ , it is lower. The kinetic pressure, density, and  $\beta$  factor rise when the test gas is injected into the core of the plasma column. This is followed by a drop in magnetic pressure, which compresses the plasma column and causes it to raise its temperature (refer to Eq. 9) [35]. The temperature of the plasma will drop as a result of thermalization brought on by collisions between ionized particles and neutral gas. The radial positions of the CS and SF at various  $\beta$  factor values



**Fig. 8** The simulated discharge current  $I_{dis}$  at different  $\beta$  factor values for 22.5 kV

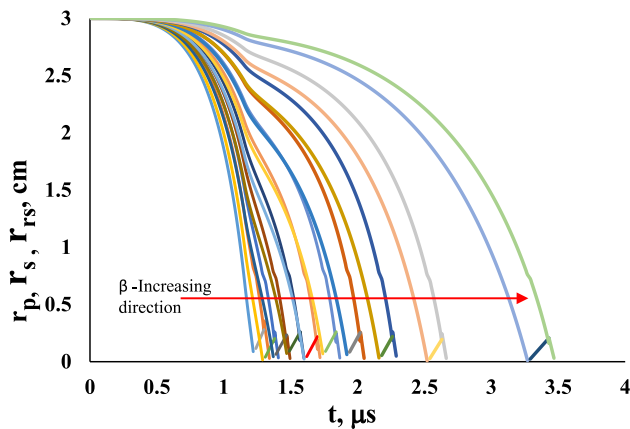


**Fig. 9** The plasma velocity  $V_p$  at different of the  $\beta$  factor values for 22.5 kV



**Fig. 10** The simulated plasma temperature  $T$  at different  $\beta$  factor values for 22.5 kV

are shown in Fig. 11. The shock wave hits the axis at different times, requiring more time for the CS to reach the pinch point [16, 38]. By raising the  $\beta$  factor and gas puff simultaneously, the CS requires a longer time to reach the



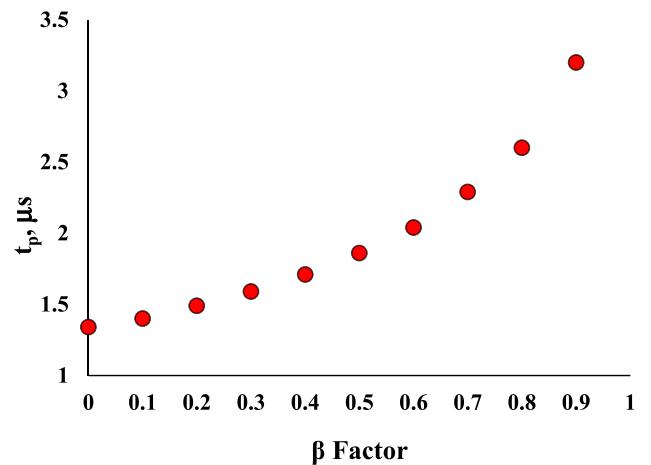
**Fig. 11** The radial position of plasma  $r_p$ , shock  $r_s$  and reflected shock wave  $r_{rs}$  at different  $\beta$  factor values for 22.5 kV

pinch point. The  $\beta$  factor is raised by injecting neutral particles into the plasma column, which raises internal pressure. As seen in Fig. 11, the shock wave is then reflected to meet the current sheath at various times.

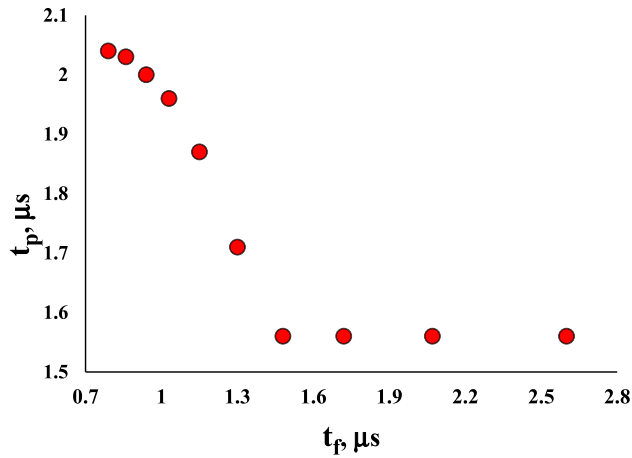
Figure 12 illustrates the pinching time  $t_p$  using a gas puff pinch mechanism with  $\beta$  factor variation. The amount of time the plasma column needs to form and compress is known as the pinching time. The figure shows the pinching time  $t_p$  is increased by increasing the  $\beta$  factor ( $\beta = P_{kr}/P_{Br}$ ), where the kinetic pressure depends on the density according to the relation “ $P_{kr} = -NkT$ ”, so the kinetic pressure and the plasma mass are increased when the plasma is injected by test gas. This causes the plasma velocity to slow down and requires more time to pinch and compress [35].

The relationship between the pinching time ( $t_p$ ) and the gas puff time ( $t_f$ ) is depicted in Fig. 13. The moment test gas, or neutral gas, begins to inject into the center of the plasma column is known as the “gas puff time”  $t_f$ . The figure illustrates how raising the puff time  $t_f$  decreases the pinching time. When the plasma is injected early by neutral particles, it takes longer for it to ionize the original gas, test gas, and form the CS since the plasma is not yet formed at the beginning of the discharge. But after a long period of time,  $t_f$ , the original gas is fully ionized; thus, it takes less time for the injected gas to ionize and compress. The relationship between the plasma velocity and the  $\beta$  factor before and after gas puff is explained in Fig. 14. The graphic shows that increasing the  $\beta$  factor with and without a gas puff reduces the plasma velocity. Since the  $\beta$  factor and injected gas are factors that affect kinetic pressure, injecting the test gas into the core of the plasma causes its mass to become heavier and its speed to become slower.

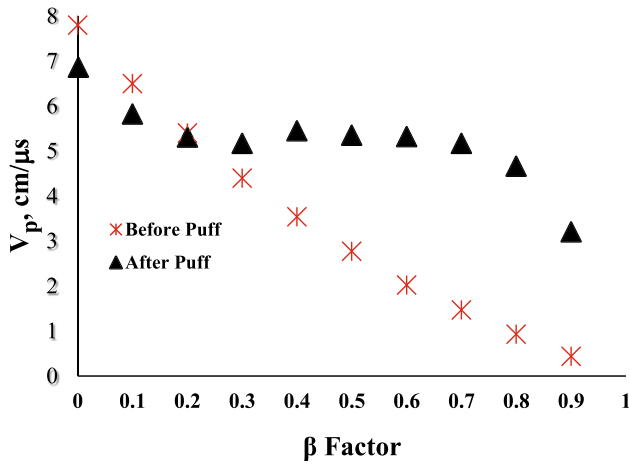
The velocity and temperature of the plasma are plotted against the time at different gas puff times  $t_f$  in Fig. 15.



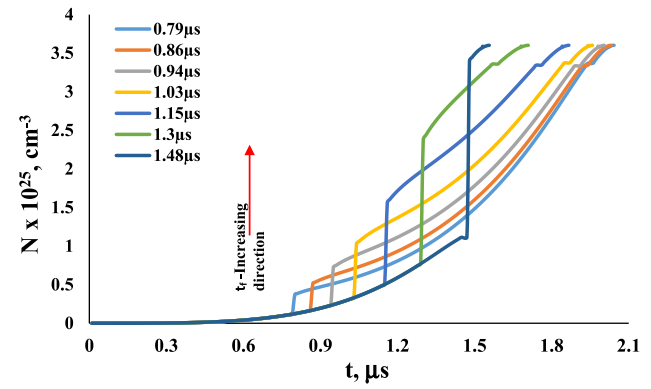
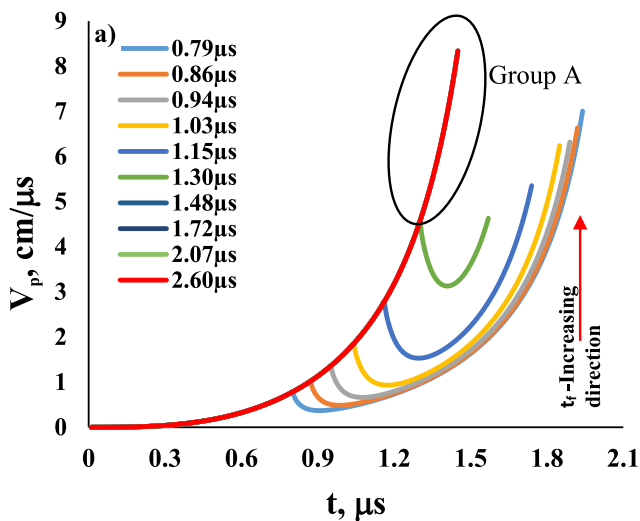
**Fig. 12** The pinching time  $t_p$  versus the different values of  $\beta$  factor for 22.5 kV



**Fig. 13** The pinching time  $t_p$  versus the injection time  $t_f$  of gas puff for 22.5 kV

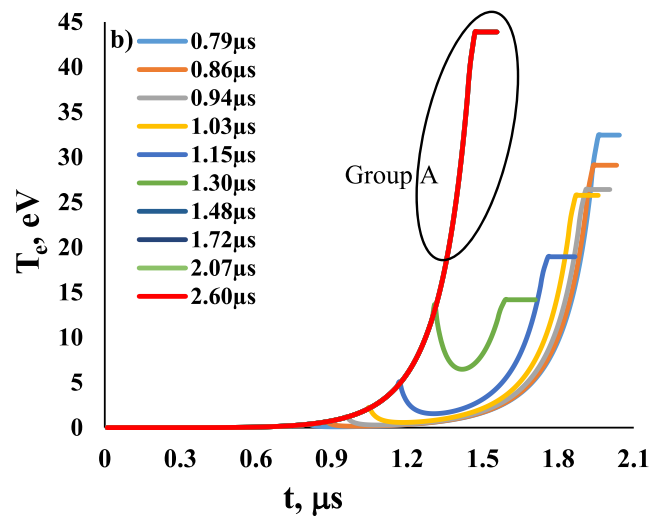


**Fig. 14** The plasma velocity  $V_p$  versus the different values of  $\beta$  factor for 22.5 kV



**Fig. 16** The plasma density ( $N$ ) as a function of discharge time at different times ( $t_f$ ) of injected gas

Over time, both the velocity and temperature increase. However, when neutral gas molecules are injected, the velocity and temperature decrease as a result of the increased plasma mass and slower plasma velocity. Additionally, a thermalization process between the neutral and ionized particles occurs, causing the plasma to cool. Following the injection of neutral particles, which ionize and compress radially, the temperature and velocity start to rise once more. As previously mentioned, Fig. 15 illustrates how temperature and velocity are impacted at a small gas puff time  $t_f$ . However, at a great time  $t_f$ , it is evident that they exhibit the same values and behavior (Group A: includes  $V_p$  and  $T_e$  curves for  $t_f = 1.48, 1.72, 2.07, 2.61 \mu s$ ) because the created, compressed, and its particles plasma column are fully ionized; hence, the injected neutral gas ionizes fast without influencing the plasma's temperature or velocity. In the current paper, the effect of the gas puff at late times of  $t_f$  disappears since the radiative phase (slow



**Fig. 15** a The plasma velocity  $V_p$  and b the temperature  $T_e$  as a function of discharge time at different times  $t_f$  of injected gas



compression phase) is not incorporated in the current model. In Fig. 16, the plasma density is shown versus the time at different gas puff time's  $t_f$ . In general, when the test gas is injected into the plasma core, the density often increases suddenly. However, because the injected gas is ionized by both the shock wave and ionized particles inside the plasma column, the density increases abruptly and sharply at large gas injection times, here equal to 1.47  $\mu$ s.

## Conclusion

The gas puff pinch device that would be created and used in the Magnetic Pinch lab. EAEA was simulated by using a snowplow model. The effect of the kinetic pressure as a function of the  $\beta$  factor on the dynamics of the plasma sheath inside the gas puff pinch device was studied using a snowplow model. Two gases were used: the original gas (first gas), which has a slow velocity, was used to form the plasma column, whereas the test gas (second gas) was only

utilized to inject at a fast velocity inside the plasma core. During testing of the model, both with and without a gas puff injection, it was evident how the gas puff affected the plasma properties. Gas particle injection results in a rise in plasma mass where the thermalization process of the neutral and ionized particles lowers the plasma's temperature and velocity while raising its density and radial position. Increasing the  $\beta$  factor results in a rise in the density and radial position of plasma, but a drop in its velocity and temperature, according to the simulated data, which indicated that the  $\beta$  factor ranged from 0 to 0.9. The pinching time  $t_p$  is decreased with the gas puff time  $t_f$ , while it is increased with the  $\beta$  factor, which depends on the kinetic pressure and gas puff.

## Appendix

See Table 2.

**Table 2** The symbols of parameters in the model

| Symbol   | Description                               | Symbol          | Description                      | Symbol                   | Description                                                              |
|----------|-------------------------------------------|-----------------|----------------------------------|--------------------------|--------------------------------------------------------------------------|
| $C_0$    | Capacitance of capacitor                  | $M$             | Plasma mass                      | $P_{Br, kr}$             | Magnetic and kinetic pressures                                           |
| $V_0$    | Charging voltage                          | $dr_p/dt = v_p$ | Radial plasma velocity           | $\gamma$                 | Specific heat ratio for ideal gas                                        |
| $L_0$    | Circuit Inductance                        | $V_1$           | Volume of the initial gas        | $V_s$                    | Shock wave velocity                                                      |
| $L_p$    | General plasma inductance                 | $V_2$           | Volume of the second (test) gas  | $t$                      | General time                                                             |
| $L_{p1}$ | Plasma inductance before the gas injected | $f_c$           | Current factor                   | $t_p$                    | Pinching time                                                            |
| $L_{p2}$ | Plasma inductance after the gas injected  | $f_m$           | Fraction of mass swept           | $t_f$                    | Injection time of test gas                                               |
| $\mu$    | Permeability of the medium                | $I_z$           | Discharge current                | $M_g$                    | Gas molecule weight                                                      |
| $z_0$    | Distance between the two electrodes       | $B_0$           | Induced magnetic field           | $R_0$                    | Universal gas constant                                                   |
| $r_0$    | Electrode radii                           | $F_{Br}$        | Major magnetic force             | $D$                      | Departure coefficient                                                    |
| $r_c$    | Circuit resistance                        | $F_{kr}$        | Kinetic force                    | $D_N$                    | Gas dissociation number                                                  |
| $r_p$    | General plasma radius                     | $F_{sr}$        | Shock wave force                 | $Z_{eff}$                | Effective charge state                                                   |
| $r_1$    | Plasma radius before the gas injected     | $F_{Fr}$        | Total force                      | $T_e$                    | Plasma temperature                                                       |
| $r_2$    | Plasma radius after the gas injected      | $P_{Br}$        | Magnetic pressure                | $N$                      | Plasma density                                                           |
| $r_s$    | Shock wave radius                         | $P_{kr}$        | Kinetic pressure                 | <b>Group<sub>1</sub></b> | Shock wave and plasma radii before injected gas                          |
| $r_{rs}$ | Reflected shock wave radius               | $P_{sr}$        | Shock wave pressure              | <b>Group<sub>2</sub></b> | Shock wave and plasma radii after injected gas                           |
| $\rho_1$ | First gas density (primary gas)           | $A_p$           | Compressed plasma area           | <b>GroupA</b>            | Includes $V_p$ and $T_e$ curves for $t_f = 1.48, 1.72, 2.07, 2.61 \mu$ s |
| $\rho_1$ | Second gas density (test gas)             | $\beta$         | Ratio of $\frac{P_{kr}}{P_{Br}}$ |                          |                                                                          |

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**Declarations**

**Conflict of interest** The author declare that he have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

**Ethical approval** Not applicable.

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